

CM3FR: Enabling NLoS THz Links and Localization Through Material Classification

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Abstract—Terahertz-band (0.1-10THz) communication is expected to play a crucial role in the next-generations of wireless technology by enabling high data-rates, ultra-secure transmission, and high-resolution sensing. However, THz-waves are susceptible to blockage, motivating non-line-of-sight (NLoS) links as a way to increase the reliability of the link. Channel-aware connections are necessary to increase reliability as the THz channel is extremely low rank. We present CM3FR, a physics-aware material classification and receiver-based localization model where 1D-CNN-ResNet architecture is utilized to handle multitask heads in a unified pipeline. Unlike existing methods that rely on dedicated radar systems, our architecture fully depends on the multipath profile of reflected signals from surrounding materials in THz links and their corresponding characteristics. A 65% accuracy in material classification and 86% accuracy in distinguishing good and poor reflectors with absolute error of 0.78m in localization over 8m is achieved, considering different transmission powers and angles-of-incidence.

Index Terms—NLoS THz channel, THz sensing, material classification, integrated sensing and communications.

I. INTRODUCTION

The terahertz (0.1 - 10 THz) band is expected to play a key role in 6G and beyond wireless technologies by enabling high data rate links, precise sensing, and enhanced security [1]. Over the past two decades, substantial research has been conducted both theoretically and experimentally to understand the unique characteristics of the channel [2], [3], [4], enhance device performance [5], and demonstrate multi-kilometer static line-of-sight (LoS) links [6]. However, maintaining a LoS link is not always feasible, as the THz link is highly susceptible to blockage [7] due to the non-negligible photon energy of THz radiation (few milli-electronvolts) that results in strong light-matter interactions and, thus, absorption, reflection, and scattering. Consequently, it is essential to rely on the NLoS component for effective communication, particularly in the context of mobile links. The main challenge in establishing NLoS links is the sparse nature of the THz channel [8], and a lack of NLoS components with sufficient signal power for reliable link setup. This complicates the identification of suitable NLoS links using 5G-type beam sweeping, resulting in increased latency and reduced throughput [9].

Although intelligent reflecting surfaces (IRSs) can help in increasing the channel rank [10], ultimately, there needs to be precise channel estimation techniques. From this perspective, channel-aware [11] NLoS link establishment can be beneficial in maintaining link reliability, laying the ground for integrated

sensing and communication (ISAC) [12]. To fully capitalize on this potential, it is crucial to understand how THz waves interact with various building materials, accounting for reflection losses and multipath effects. Special emphasis should be placed on specular reflection, as components that are diffracted and scattered generally exhibit much lower power [13], [14]. It is crucial to classify these materials within the communication environment according to their reflected multipath profiles (MP). This classification can help us establish the best links through NLoS. In fact, several studies have explored machine learning (ML) and deep learning techniques for mmWave radar-based material classification [15] and NLoS-based localization [16]; however, these studies do not adequately address the peculiarities of the THz band [17], [18]. Furthermore, a study on material classification using convolutional neural network (CNN) techniques at the sub-THz range considers reflection and penetration losses [19], making it more suitable for sensing applications and posing challenges for use in communication and ISAC contexts. There is a significant gap in research on two key aspects: i) characterizing non-line-of-sight (NLoS) links in terms of multipath effects, and ii) classifying materials and localizing receivers based on NLoS reflections, without relying on radar-like time stamping.

The contributions of this paper, based on the identified problem statements, are as follows: **i)** We characterize the NLoS link based on reflection loss and MP across different distances and incident angles, extracting important metrics crucial for understanding different building materials' performance in terms of reflection. These metrics are used as input for the physics-aware AI/ML model. **ii)** We developed a tailored multitask model based on a 1D-CNN ResNet architecture [20], designed to operate on NLoS MP and its characteristics, which performs material classification (metal, glass, drywall, and wood), assesses materials based on the MP of reflected signals, and estimates the distance between the transmitter (Tx) and receiver (Rx) without any type of synchronization. **iii)** The model uses a unified pipeline and improved architecture to minimize interference between output heads for better performance while remaining lightweight for real-time use.

The paper is organized as follows. In Sec.II, we statistically characterize NLoS links by extensive experimental data capture. In Sec.III, we discuss our CNN-based tailored unified multitasking model. In Sec.IV, we present the results for our CNN-based ML model output heads. Finally, in Sec.V, we summarize our work and draw conclusions.

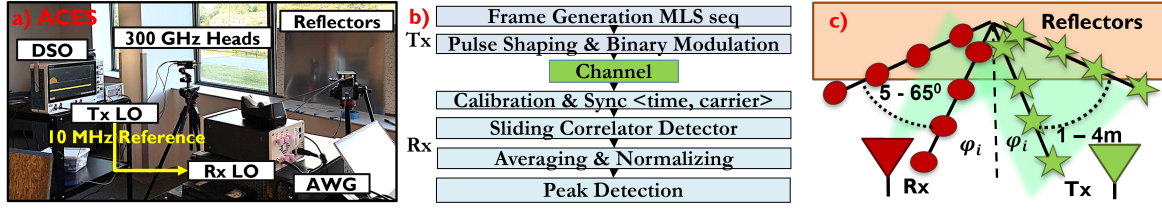


Fig. 1. a) The ACES testbed and channel sounding environment, b) Channel sounder signal processing backend and c) NLoS channel sounding strategy.

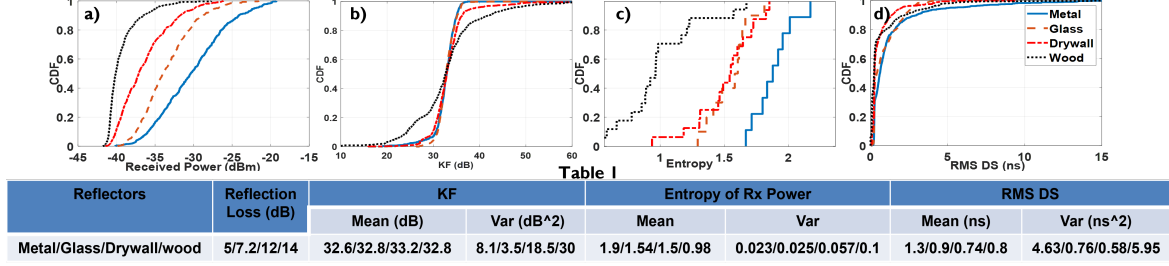


Fig. 2. CDF plot: a) Rx power, b) KF, c) Entropy of Rx power, and d) RMS DS. Table represents mean and variance of each metric for different materials.

II. NLOS STATISTICS

In this section, we analyze the characteristics of specularly reflected NLoS links by i) conducting a channel sounding campaign and ii) examining the NLoS link characteristics.

A. Channel Sounding Methodology

We conducted channel sounding measurement using the Advanced Communication, Electronics, and Sensing (ACES) [21] testbed with 26 dBi antennas and 4GHz bandwidth at 300 GHz to capture MPs for specularly reflected signals from various materials commonly used in building construction, such as metal, glass, drywall, and wood. This testing occurred in a conference room environment, incorporating the effects of the real scenario (Fig. 1(a)). To capture the MP statistics comprehensively, we considered different Tx and Rx separations from the reflector, varying from 1 m to 4 m, creating 16 distinct configurations and incident angles (5°, 25°, 45°, and 65°) to assess the effects of surface roughness and the diffusely scattered signals [22] (Fig. 1(c)). Further, to account for antenna misalignment, different transmit powers (8, 5, 2, -1, and -4 dBm) are utilized. We collected data three times for each configuration, resulting in a total of 3,840 MPs (4 × 16 × 4 × 3 × 5). For this purpose, we developed a sliding correlator-type channel sounder (Fig. 1(b)) with a customized signal processing backend designed to meet THz band requirements with a resolution of 0.25 ns and a dynamic range of 60 dB [8].

B. NLoS Link Characteristics

We analyze the NLoS link using the metrics presented in Fig. 2 Table 1. Our goal is to understand the reflection MP characteristics of various materials and to develop a physics-aware multi-task ML model that can effectively perform classification and localization.

The Rx power after reflection from various materials is shown in the cumulative distribution function (CDF) plot in Fig. 2 (a). Metal reflects the most power in the specular direction, followed by glass, drywall, and wood, due to their smoother surfaces, resulting in a less scattered signal. This

is supported by the reflection loss data in Fig. 2 Table 1. The K-factor (KF), shown in Fig. 2 (b), compares the power from the first-order reflection ($P_{1st-Ref}$) component to the scattered and other reflection components ($2S^2$) and is given by $K = P_{1st-Ref}/2S^2$ and $10\log_{10}(K)$ in dB scale. The mean values of KF are similar across all materials; the variance is lower for metal and glass, while it is higher for drywall and wood. The main reason for having a nearby KF for the NLoS link is that materials like metal and glass have low reflection loss, allowing signals to reflect off nearby objects, leading to higher-order reflections and a reduced KF. In contrast, drywall and wood show high variance due to their rough surfaces, which scatter signals more at higher angles and weaken the main reflection. Taking into account reflection loss and the KF, we can conclude that drywall and wood can provide a reliable link for a receiver if positioned nearby. On the other hand, metal and glass are better options for longer distances. Therefore, our machine learning model classifies metal and glass as good reflectors and drywall and wood as poor reflectors.

Additionally, for sensing and material classification, relying solely on the RX power after reflection can result in suboptimal algorithm performance, as this power is affected by antenna alignment and other factors. Thus, we emphasize metrics that more accurately reflect the multipath structure. In this regard, we evaluate the entropy of the reflected RX power associated with different peaks in MP (Fig. 2 (c)). This is computed using Shannon's entropy formula:

$$H = - \sum_{n=1}^N p(n) \ln p(n), \quad (1)$$

where $p(n)$ is the histogram-based probability distributions of powers in a particular MP, and N is the number of bins in the histogram. This approach measures the expected (i.e., average) amount of information conveyed by the randomness of the power profile of the MP. For a metal surface, entropy is high with low variance, which is similar to the explanation for reduced KF. This causes increased randomness in the Rx

power due to fading, and the entropy decreases for glass, followed by drywall and wood. For a similar reason, the root mean square (RMS) delay spread (DS) (Fig. 2(d)) for metal is higher compared to glass, drywall, and wood, given by $\tau_{\text{RMS}} = \sqrt{\sum_i (d_i - \hat{d})^2 p_i^2 / \sum_i p_i^2}$, where d_i and p_i are the delay and received power of propagation path i , respectively, and $\hat{d}$ is the mean delay represented by $\hat{d} = \frac{\sum_i d_i p_i}{\sum_i p_i}$.

III. MULTI-TASK FEATURE EXTRACTION MODEL

A. Model Architecture

To carry out multiple tasks, such as performing binary classification (2-Class) of good and poor reflectors, categorizing materials into four groups (4-Class - metal, glass, drywall, and wood), and localizing the Rx by reflection path, we developed a unified model - the CNN-based Multi-tasking 3-Stage Fusion Residual Model (CM3FR).

a) Backbone: Our multi-task model utilizes a 1-D CNN ResNet-18 [20] backbone core consists of residual blocks, which allows for deep stacking without the vanishing gradient problem, ensuring that the model learns effectively without slow adaptation to achieve the first convergence. The backbone is responsible for extracting multi-scale features from the input data and feeding them into the task-specific heads (Fig. 3). This approach avoids multiple models and repeated pre-processing, keeping the runtime and energy footprint low, which is important for on-device, latency-sensitive operations.

In particular, each task-specific head receives three different views of the signal (see Fig. 3): i) Multi-scale features from the backbone core, as previously mentioned; ii) Separate supporting task-specific stems that employs a 3-layer shallow CNN applied directly to the MP; and iii) an RMS DS embedded learning layer. Keeping task-specific stems shallow reduces latency on constrained devices and prevents the model from overfitting on the training data. Keeping them separate reduces feature interference from other heads. If everything is optimized jointly from the start, dominant tasks can overshadow weaker ones. Within each head, these features are concatenated and passed through a fusion Multi-Layer Perceptron (MLP) [23] that rebalances the scales before the final output. Fusion is performed for each task-specific head, allowing each task to emphasize the features it requires. The RMS DS embedding provides support by adding features to classify the materials, which have close power profiles, such as glass and metal.

b) Multi-Head Output: The network branches into four prediction heads: i) *2-Class Head:* This head focuses on Binary Material Classification, distinguishing between “good” (metal/glass) and “poor” (wood/drywall) reflectors. ii) *4-Class Head:* This head categorizes different material types. iii) *Entropy Head:* This head predicts multipath power entropy, which aids in extracting physics-aware features. iv) *Localization Head:* This head estimates the total path distance from the Tx to the Rx through the reflector based on MP features.

B. Training Process:

a) 3-Stage method: We train the model in three stages, under a unified pipeline. The training schedule is illustrated

in Fig. 3. *(i) Warm-up:* In this initial stage, we train the entire network (backbone core with supporting stems) to establish a shared representation of the tasks. *(ii) Task-specific stem specialization:* Next, we freeze the backbone core and the supporting RMS DS embedding and fine-tune each task-specific stem head at a time using 3-layer shallow CNN models to keep it lightweight. *(iii) Retune:* Finally, we briefly unfreeze all components and conduct a joint fine-tuning process. For retune, a lower learning rate is used to ensure conservative, stable updates that align with head-specific adaptations.

b) Training Optimization: The loss functions for various tasks provide insight into their performance. We use binary cross-entropy (BCE) for binary classification, entropy employs mean squared error (MSE), and 4-class classification and localization use mean absolute error (MAE). In a multi-task model, it is vital to consider the combined impact of each task on the loss function by assigning appropriate weights. Careful adjustment of these weights ensures that no specific task is favored over others. To balance the weights associated with these task losses and to get the cumulative task loss, we utilize the method proposed by Kendall & Gal [24] as,

$$\mathcal{L}_{\text{MTL}} = \sum_{i=1}^T \left(e^{-2s_i} \mathcal{L}_i + s_i \right), \quad s_i \in \mathbb{R} \quad (2)$$

where $\mathcal{L}_i$ denotes the loss associated with the i^{th} task and $s_i \in \mathbb{R}$ is a learnable scalar representing the task-dependent uncertainty. The term e^{-2s_i} serves as an adaptive weighting factor that automatically scales each task’s loss contribution based on its relative uncertainty; tasks with higher uncertainty (larger s_i) are down-weighted, preventing them from dominating the shared backbone gradients. We employ the AdamW optimizer [25] to optimize the weights with Kendall and Gal method, utilizing a low learning rate, as previously discussed. Specifically, we use a learning rate of 1×10^{-4} during the warm-up phase, 1×10^{-4} for task-specific head specialization, and 1×10^{-5} for retuning. To efficiently process data in a constrained unit, we use batching and shuffling. We group 3,840 MPs into batches of 16 MPs and shuffle them to avoid overtraining on any single batch.

c) Training Criterion: We use BCE as our loss function for 2-Class classification, which quantifies the difference between the predicted probabilities of our model and the actual binary labels (0 or 1, representing good or poor reflector). This loss function also aligns well with the decision metrics we report, such as accuracy and confusion matrix values [26]. The BCE is given by,

$$\text{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)], \quad (3)$$

where y_i represents the true label and p_i denotes the predicted probability. For the regression head related to entropy, we use MSE Loss, denoted as $\mathcal{L}_{\text{MSE}}$, which is defined as follows,

$$\mathcal{L}_{\text{MSE}}(\hat{\mathbf{H}}, \mathbf{H}) = \frac{1}{N} \sum_{n=1}^N \left(\hat{H}_n - H_n \right)^2$$

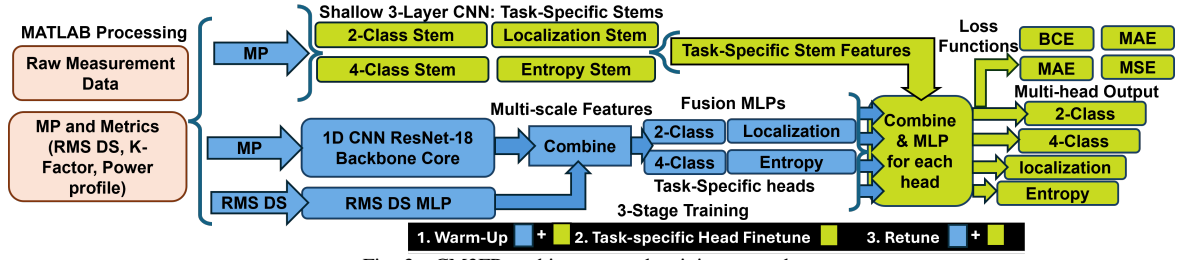


Fig. 3. CM3FR architecture and training procedure.

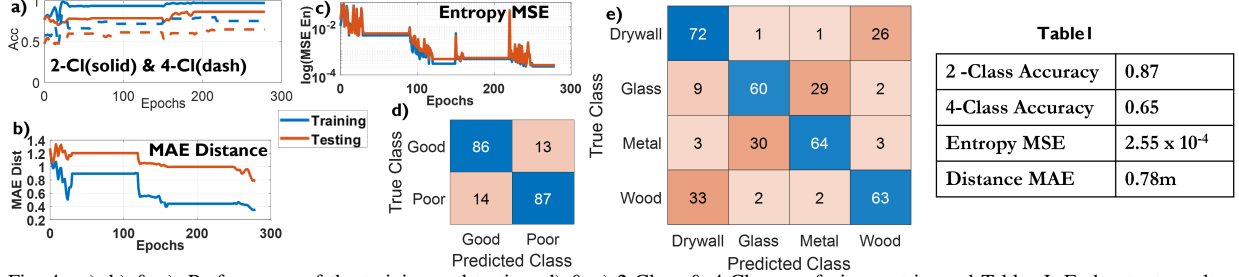


Fig. 4. a), b) & c): Performance of the training and testing, d) & e) 2-Class & 4-Class confusion matrix, and Tables I: End output results.

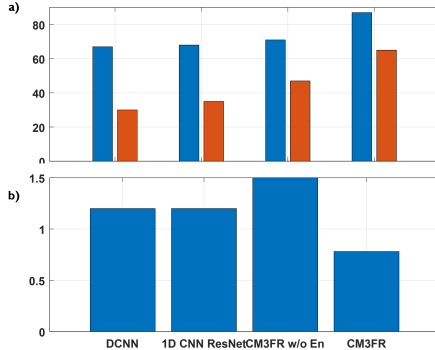


Fig. 5. Comparison of the output results between different models: DCNN, the 1D-CNN ResNet backbone, CM3FR without the entropy head, and CM3FR.

In this equation, $\hat{H}_n$ represents the predicted value, while H_n denotes the actual value of the entropy, as shown in Equation 1. The MSE loss, known as L_2 criterion [26], imposes higher penalties for larger prediction errors. This is especially useful for addressing significant errors in predicting entropy, helping to guide decision-making more accurately. We use MAE for 4-Class classification and localization regression model, as

$$\mathcal{L}_{\text{MAE}}(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{N} \sum_{n=1}^N |\hat{y}_n - y_n|$$

Here, $\hat{y}_n$ represents the predicted value, while y_n denotes the actual value. We select MAE, or L_1 loss, for both localization and 4-class classification regression due to its ability to handle overlapping data. For example, an MP from a strong reflector at low power and greater distance could exhibit similar properties to a poor reflector with high power at a closer distance.

IV. RESULTS

A. Training and Testing Performance

We split the data, with 85% used for training and 15% for testing. We evaluate the model's performance using

the following metrics: binary prediction accuracy and 4-class prediction accuracy, entropy-based MSE, and distance-based MAE, based on the final checkpoint from the re-tuning stage. Accuracy is calculated as follows: $\text{Acc} = \text{Number}_{\text{correct-predictions}} / \text{Number}_{\text{total-events}}$. Fig. 4 illustrates the training and testing performance with increasing epochs. The performance metrics during training and testing closely align with each other, indicating that our model has converged well and shows good accuracy in predicting task-specific heads. It is evident from the Fig. 4(a), (b), and (c) that the warm-up stage alone does not significantly enhance performance metrics. However, after freezing the layers and applying the head-specific stem extraction, we observe a notable improvement, evident as a jump, which supports our model design. However, after freezing the layers and applying stem extraction, we notice a significant improvement, evident as a jump, that validates our model design.

B. Task-specific Results

a) **Classification:** The end results of our model is summarized in Table I (Fig. 4). The accuracy for 2-class and 4-class classification is 0.87 and 0.65, with F1 scores of 0.86 and 0.65, respectively. In both cases, we obtain F1 score ≥ 0.65 , which indicates a good balance of precision and recall, i.e., both low false positives (predicted positive but actually negative) and low false negatives (missed positives). The performance indicated that accurate classification of the material is possible through reflection only, which makes our physics-aware model suitable for implementation in communication and ISAC scenarios. In Fig. 4(d) and (e), we present the confusion matrices for both binary and four-class classifications of our model. It indicates that while the accuracy for 2-Class classification is good, the lower accuracy observed in the 4-class classification is primarily due to confusion between the *glass* $\leftrightarrow$ *metal* and *drywall* $\leftrightarrow$ *wood*.

b) **Localization:** We achieved a MAE of 0.78m in estimating the distance between Tx and Rx using only NLoS MP,

without a radar-type reference signal. This suggests accurate localization of the receiver is possible, as we can obtain angular information through direction-finding algorithms [27]. This level of accuracy can be advantageous for tracking the receiver or mobile units within the environment. The accuracy can be further improved by using higher bandwidth and antenna gain. Additionally, we obtained an MSE of 2.55×10^{-4} for the entropy head, demonstrating that we successfully developed a physics-aware model that achieves improved performance.

c) Comparison: we compare the accuracy of 2-class, and 4-class classification, and MAE of localization in Fig. 5 (a) and (b) with a DCNN [15], 1D-CNN ResNet backbone core alone, CM3FR without the entropy head, and CM3FR. It is evident that CM3FR outperforms the DCNN by a significant margin. Furthermore, the results indicate that the 3-stage tuning and task-specific stems contribute to improved outputs.

V. CONCLUSION

In this paper, we statistically estimated crucial metrics associated with the NLoS MP by capturing experimental data at 300 GHz using the ACES testbed. We developed a unified physics-aware CNN-based model, CM3FR, to classify different building material types and localize Rx in an ISAC environment tailored for THz-band systems. The model achieves 65% accuracy for classifying four types of materials and 87% accuracy for binary classification. This is accomplished using multipath information from the reflected signal, without any radar-based reference signaling. Further, the model provides an MAE of 0.78 m for Rx location estimation over an 8 m NLoS link. This functionality can be utilized for device tracking, provided that at least one NLoS link exists between the Tx and Rx. The performance could be enhanced by using higher bandwidth and a high-gain antenna setup.

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